

Data Engineering Anomaly Detection Explanation Generator

GenAI-powered system to detect anomalies in data pipelines and automatically generate root cause explanations and remediation actions.

Synthetic logs + inspired test datasets.



Problem Statement

Modern data pipelines generate massive logs and metrics.

When failures occur:

Manual Inspection

Engineers manually inspect logs

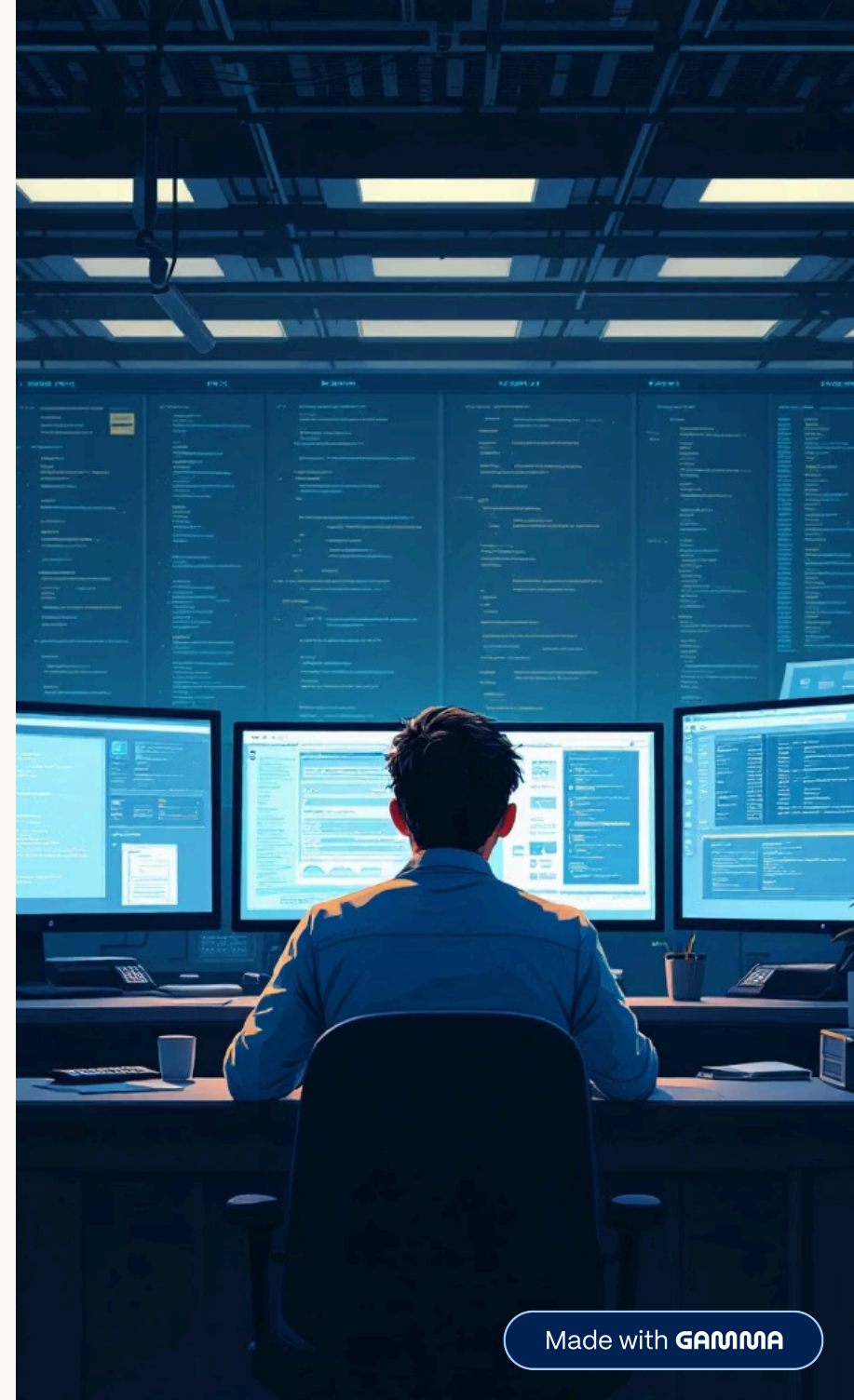
Slow Investigation

Root cause investigation takes hours

Delayed Resolution

Incident resolution is delayed

- ❏ Organisations need **automated anomaly detection and explanation systems.**



Solution

Our system provides an **AI-powered anomaly investigation assistant**.

Capabilities:



Parse Pipeline Logs

Parse pipeline logs



Analyse Pipeline Metrics

Analyse pipeline metrics



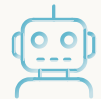
Detect Anomalies Using ML

Detect anomalies using ML



Retrieve Historical Incidents Using RAG

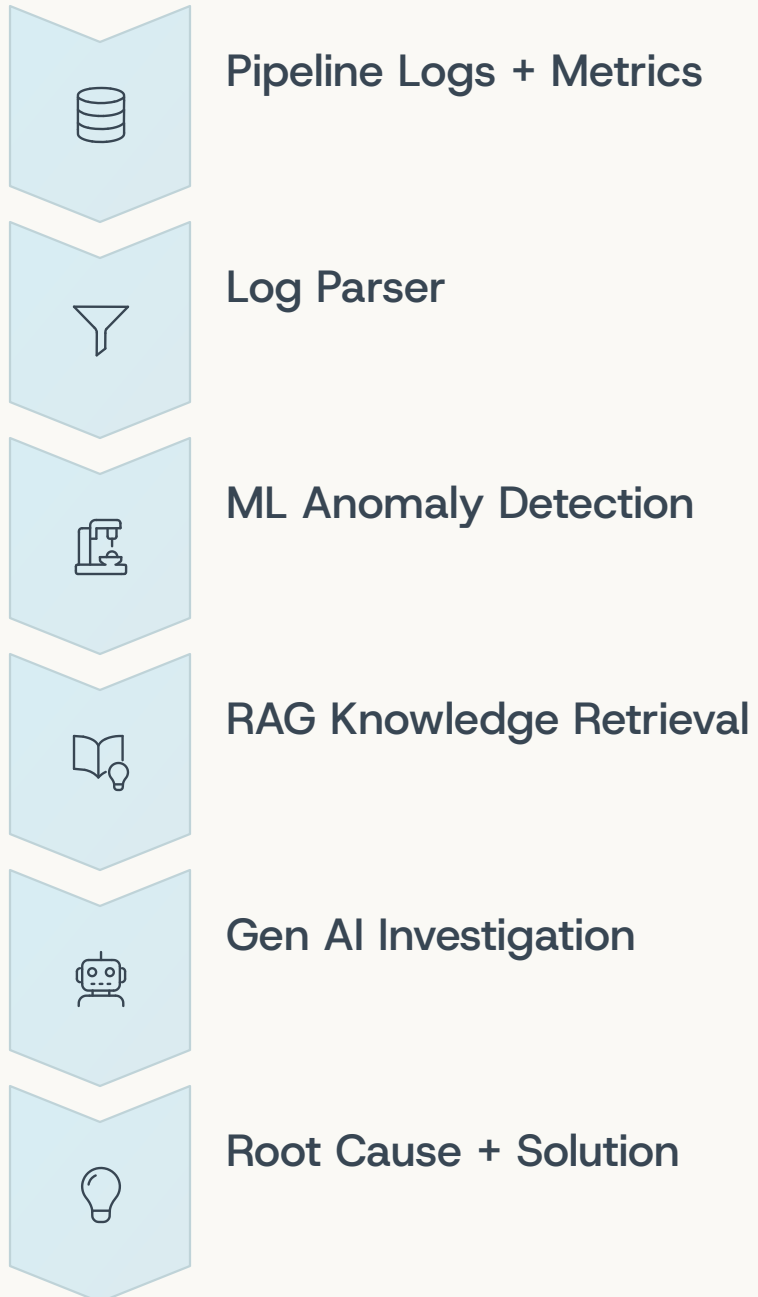
Retrieve historical incidents using RAG



Generate Root Cause Explanations Using LLM

Generate root cause explanations using LLM

System Architecture



Synthetic Dataset Generation

To simulate realistic scenarios we generated:

5000

Synthetic Pipeline Logs

Generated to simulate realistic scenarios

Each log contains:

Pipeline Name	Stage
Latency	Processed Records
Execution Status	Error Messages

Anomaly Patterns Simulated

The dataset includes multiple anomaly scenarios:

1

Latency Spikes

2

Low Records Processed

3

Upstream API throttling Failures

4

Ingestion Failures

5

Infrastructure Performance Issues



These patterns simulate **real-world data pipeline incidents**.

Example Pipeline Log

A sample log entry from the synthetic dataset:

```
{  
  "timestamp": "2026-03-05 09:50:00",  
  "pipeline": "orders_etl",  
  "stage": "ingestion",  
  "latency": 115,  
  "records_processed": 1001500,  
  "status": "SUCCESS"  
}
```




Example Anomaly

An anomaly detected in pipeline logs:

Records Processed

Dropped from 1,000,000 to 12,000

Runtime

Increased significantly

Error Rate

Increased

CPU Usage

Spiked

📋 The system flags this as a **critical pipeline anomaly**.

AI Generated Explanation

Root Cause

Upstream API ingestion failure caused incomplete data extraction.

Impact

Downstream transformation stage failed due to insufficient input data.

Recommended Solution

01

Validate Upstream API Health

Validate upstream API health

02

Restart Ingestion Pipeline

Restart ingestion pipeline

03

Add Monitoring Alerts

Add monitoring alerts

Technologies Used

Core technologies in the solution:



Large Language Models



Machine Learning Anomaly Detection



Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation (RAG)



Synthetic Dataset Generation



Streamlit Dashboard

Streamlit dashboard

Future Enhancements

Agentic AI

To take corrective actions based on anomalies detected

Real time logs processing

Can take different kinds of inputs